MATH2601 – Higher Linear Algebra

# Topic 4 – Inner product spaces
# Lecture 4.2 – Orthogonality in inner product spaces

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# Orthogonality

Let $(V, \mathbb{F})$ be an inner product space with inner product $\langle \cdot \rangle$. (Recall $\mathbb{F}$ represents only $\mathbb{R}$ or $\mathbb{C}$ for this section.) Then we have the following generalisations of orthogonality.

**Definition.** Two vectors $\mathbf{u}, \mathbf{v} \in V$ are orthogonal if and only if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

**Definition.** An orthogonal set is any set $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\} \subseteq V$ whose elements are pairwise orthogonal, that is $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ for all $i \neq j$.

**Definition.** An orthogonal set $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\} \subseteq V$ is orthonormal if every element has norm 1, that is $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$ for all $i, j$.

**Fact.** Any orthogonal set of nonzero vectors is linearly independent in $V$.

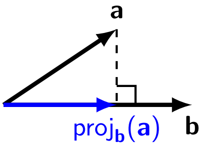
**Example.** Recall the inner product space $M_{2,2}(\mathbb{F})$ with inner product $\langle A, B \rangle = \mathrm{tr}(A^* B)$ for all $A, B \in M_{2,2}(\mathbb{F})$. The standard basis $\{E_{1,1}, E_{1,2}, E_{2,1}, E_{2,2}\}$ is an orthonormal basis for this inner product space.

Another nonstandard orthonormal basis is below. Exercise: Check these!

$$\left\{ \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \right\}.$$

# Vector projection

Recall that for geometric vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^2$, the projection from $\mathbf{a}$ onto $\mathbf{b}$ is $\mathrm{proj}_{\mathbf{b}}(\mathbf{a}) = \dfrac{\mathbf{b} \cdot \mathbf{a}}{|\mathbf{b}|^2}\mathbf{b}$.




We can adapt this construction for general inner product spaces.

**Definition.** Given an inner product space $(V, \mathbb{F})$ with inner product $\langle \cdot \rangle$ and $\mathbf{u}, \mathbf{v} \in V$ with $\mathbf{v} \neq \mathbf{0}$, the projection of $\mathbf{u}$ onto $\mathbf{v}$, denoted $\mathrm{proj}_{\mathbf{v}}(\mathbf{u})$, is given by $\mathrm{proj}_{\mathbf{v}}(\mathbf{u}) = \dfrac{\langle \mathbf{v}, \mathbf{u} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle}\mathbf{v}$. (Note the order of vectors in the numerator.)

**Lemma.** For a fixed $\mathbf{v} \in V$, the projection mapping $\mathrm{proj}_{\mathbf{v}} : V \to V$ is linear.

**Proof.** Due to the fact the inner product is linear in the second argument, we can easily show that $\mathrm{proj}_{\mathbf{v}}(\mathbf{u} + \lambda \mathbf{w}) = \mathrm{proj}_{\mathbf{v}}(\mathbf{u}) + \lambda\, \mathrm{proj}_{\mathbf{v}}(\mathbf{w})$ for all $\mathbf{u}, \mathbf{w} \in V$ and $\lambda \in \mathbb{F}$.

**Lemma.** The vectors $\mathbf{v}$ and $\mathbf{u} - \mathrm{proj}_{\mathbf{v}}(\mathbf{u})$ are orthogonal.

**Proof.** Note $\langle \mathbf{v}, \mathbf{u} - \mathrm{proj}_{\mathbf{v}}(\mathbf{u}) \rangle = \langle \mathbf{v}, \mathbf{u} \rangle - \langle \mathbf{v}, \frac{\langle \mathbf{v}, \mathbf{u} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle}\mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle - \frac{\langle \mathbf{v}, \mathbf{u} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle}\langle \mathbf{v}, \mathbf{v} \rangle = 0.$

# Example

**Exercise.** In the inner product space $\mathbb{C}^2$ with standard inner (dot) product, find the projection of $\mathbf{u} = \begin{pmatrix} 2+i \\ -3+5i \end{pmatrix}$ onto $\mathbf{v} = \begin{pmatrix} 1+2i \\ 3+i \end{pmatrix}$.

**Solution.** We have

$$\langle \mathbf{v}, \mathbf{u} \rangle = (\overline{1+2i})(2+i) + (\overline{3+i})(-3+5i) = (4-3i) + (-4+18i) = 15i,$$
$$\langle \mathbf{v}, \mathbf{v} \rangle = (\overline{1+2i})(1+2i) + (\overline{3+i})(3+i) = -5+10 = 15.$$

So the projection of $\mathbf{u}$ onto $\mathbf{v}$ is

$$\mathrm{proj}_{\mathbf{v}}(\mathbf{u}) = \frac{\langle \mathbf{v}, \mathbf{u} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle}\mathbf{v} = i \begin{pmatrix} 1+2i \\ 3+i \end{pmatrix} = \begin{pmatrix} -2+i \\ -1+3i \end{pmatrix}.$$

**Exercise.** Find an orthonormal basis for $\mathbb{C}^2$ including a scalar multiple of $\mathbf{v}$.

**Solution.** We know $\mathbf{p} = \mathbf{u} - \mathrm{proj}_{\mathbf{v}}(\mathbf{u}) = \begin{pmatrix} 4 \\ -2+2i \end{pmatrix}$ is orthogonal with $\mathbf{v}$.

So $\{\mathbf{v}, \mathbf{p}\}$ is a linearly independent set of size $2 = \dim(\mathbb{C}^2)$, meaning it is an orthogonal basis. To create an orthonormal basis, we just have to divide each vector through by its norm (length). So a solution is

$$\left\{ \frac{1}{\sqrt{15}} \begin{pmatrix} 1+2i \\ 3+i \end{pmatrix}, \frac{1}{\sqrt{24}} \begin{pmatrix} 4 \\ -2+2i \end{pmatrix} \right\}.$$

**Lemma.** If an inner product space $(V, \mathbb{F})$ contains an orthogonal set of nonzero vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, then for any $\mathbf{u} \in \operatorname{span}(S)$ we have $\mathbf{u} = \sum_{i=1}^{n} \operatorname{proj}_{\mathbf{v}_i}(\mathbf{u})$.

**Proof.** Since $S$ is orthogonal with no zero vectors, it is linearly independent, and so there are unique scalars $\lambda_i \in \mathbb{F}$ such that $\mathbf{u} = \sum \lambda_i \mathbf{v}_i$. Notice that for some particular $k$, taking the inner product of both sides with $\mathbf{v}_k$ in the first argument gives

$$\langle \mathbf{v}_k, \mathbf{u} \rangle = \langle \mathbf{v}_k, \sum_{i=1}^{n} \lambda_i \mathbf{v}_i \rangle = \sum_{i=1}^{n} \lambda_i \langle \mathbf{v}_k, \mathbf{v}_i \rangle = \lambda_k \langle \mathbf{v}_k, \mathbf{v}_k \rangle.$$

So $\lambda_k = \dfrac{\langle \mathbf{v}_k, \mathbf{u} \rangle}{\langle \mathbf{v}_k, \mathbf{v}_k \rangle}$ for each $k$, meaning $\mathbf{u} = \sum_{i=1}^{n} \operatorname{proj}_{\mathbf{v}_i}(\mathbf{u})$.

**Corollary.** If an inner product space $(V, \mathbb{F})$ has orthonormal basis $S = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$, then for any $\mathbf{u} \in \operatorname{span}(S)$ we have $\mathbf{u} = \sum_{i=1}^{n} \langle \mathbf{e}_i, \mathbf{u} \rangle \mathbf{e}_i$,

that is, $[\mathbf{u}]_S = \big( \langle \mathbf{e}_i, \mathbf{u} \rangle \big)_{1 \le i \le n}$.

# Gram-Schmidt process

Based on the previous example, we now describe a method for producing an orthonormal basis for any finite-dimensional inner product space.

**Algorithm.** (Gram-Schmidt process)
Given a finite-dimensional inner product space $(V, \mathbb{F})$ with a basis $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, construct the following new vectors $\mathbf{w}_i \in V$:

$$\begin{aligned}
\mathbf{w}_1 &= \mathbf{v}_1, \\
\mathbf{w}_2 &= \mathbf{v}_2 - \mathrm{proj}_{\mathbf{w}_1}(\mathbf{v}_2), \\
\mathbf{w}_3 &= \mathbf{v}_3 - \mathrm{proj}_{\mathbf{w}_1}(\mathbf{v}_3) - \mathrm{proj}_{\mathbf{w}_2}(\mathbf{v}_3), \\
&\vdots \\
\mathbf{w}_n &= \mathbf{v}_n - \sum_{i=1}^{n-1} \mathrm{proj}_{\mathbf{w}_i}(\mathbf{v}_n).
\end{aligned}$$

Next, for each $1 \leq i \leq n$, set $\mathbf{e}_i = \dfrac{\mathbf{w}_i}{||\mathbf{w}_i||}$.

By construction, the set $S = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is an orthonormal basis for $V$.

**Corollary.** Every finite-dimensional inner product space has an orthonormal basis.

# Proof of Gram-Schmidt

**Proof.** Let $P(k)$ be the statement "$\{\mathbf{w}_1, \dots, \mathbf{w}_k\}$ is an orthogonal basis for $\text{span}(\{\mathbf{v}_1, \dots, \mathbf{v}_k\})$". Clearly $P(1)$ is true (where the orthogonality is true vacuously), though we can also easily confirm that $P(2)$ is true from a previous lemma.

Now suppose that $P(k-1)$ is true for some integer $k$ between 1 and $n$. Then consider $\mathbf{w}_k = \mathbf{v}_k - \sum_{i=1}^{k-1} \text{proj}_{\mathbf{w}_i}(\mathbf{v}_k)$.

For any particular $1 \le \ell < k$, taking the inner product of both sides with $\mathbf{w}_\ell$ in the first argument gives

$$\langle \mathbf{w}_\ell, \mathbf{w}_k \rangle = \left\langle \mathbf{w}_\ell, \mathbf{v}_k - \sum_{i=1}^{k-1} \frac{\langle \mathbf{w}_i, \mathbf{v}_k \rangle}{\langle \mathbf{w}_i, \mathbf{w}_i \rangle} \mathbf{w}_i \right\rangle = \langle \mathbf{w}_\ell, \mathbf{v}_k \rangle - \frac{\langle \mathbf{w}_\ell, \mathbf{v}_k \rangle}{\langle \mathbf{w}_\ell, \mathbf{w}_\ell \rangle} \langle \mathbf{w}_\ell, \mathbf{w}_\ell \rangle = 0.$$

Furthermore, by assumption $\text{span}(\{\mathbf{w}_1, \dots, \mathbf{w}_{k-1}\}) = \text{span}(\{\mathbf{v}_1, \dots, \mathbf{v}_{k-1}\})$, so over the basis $\{\mathbf{v}_1, \dots, \mathbf{v}_{k-1}, \mathbf{v}_k\}$, the unique linear combination for $\mathbf{w}_k$ has $k$th coordinate 1. So $\mathbf{w}_k \notin \text{span}(\{\mathbf{v}_1, \dots, \mathbf{v}_{k-1}\})$, meaning

$$\text{span}(\{\mathbf{v}_1, \dots, \mathbf{v}_k\}) = \text{span}(\{\mathbf{w}_1, \dots, \mathbf{w}_{k-1}, \mathbf{v}_k\}) = \text{span}(\{\mathbf{w}_1, \dots, \mathbf{w}_{k-1}, \mathbf{w}_k\}).$$

So $\{\mathbf{w}_1, \dots, \mathbf{w}_k\}$ is orthogonal and a basis for $\text{span}(\{\mathbf{v}_1, \dots, \mathbf{v}_k\})$, meaning $P(k-1) \implies P(k)$ and so the statement is true for all positive integers $k$.

# Gram-Schmidt example

**Exercise.** In the inner product space $\mathbb{C}^3$ with standard inner (dot) product, find an orthonormal basis that includes a scalar multiple of $\mathbf{v}_1 = \begin{pmatrix} 1 \\ i \\ -i \end{pmatrix}$.

**Solution.** We first build up a basis for $\mathbb{C}^3$ by adding standard basis vectors, for example $\{\mathbf{v}_1, \mathbf{v}_2 = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix}^T, \mathbf{v}_3 = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}^T\}$.

We can now apply the Gram-Schmidt process. First set $\mathbf{w}_1 = \mathbf{v}_1$. Then

$$\mathbf{w}_2 = \mathbf{v}_2 - \mathrm{proj}_{\mathbf{w}_1}(\mathbf{v}_2) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} - \frac{(-i)}{3}\begin{pmatrix} 1 \\ i \\ -i \end{pmatrix} = \frac{1}{3}\begin{pmatrix} i \\ 2 \\ 1 \end{pmatrix},$$

$$\mathbf{w}_3 = \mathbf{v}_3 - \mathrm{proj}_{\mathbf{w}_1}(\mathbf{v}_3) - \mathrm{proj}_{\mathbf{w}_2}(\mathbf{v}_3) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} - \frac{i}{3}\begin{pmatrix} 1 \\ i \\ -i \end{pmatrix} - \frac{1}{6}\begin{pmatrix} i \\ 2 \\ 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} -i \\ 0 \\ 1 \end{pmatrix}.$$

Finally we can divide each vector by its norm to get the orthonormal basis

$$\left\{ \frac{1}{\sqrt{3}}\begin{pmatrix} 1 \\ i \\ -i \end{pmatrix}, \frac{1}{\sqrt{6}}\begin{pmatrix} i \\ 2 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}}\begin{pmatrix} -i \\ 0 \\ 1 \end{pmatrix} \right\}.$$

# Gram-Schmidt example

**Exercise.** In the inner product space $(\mathcal{C}([0,1]), \mathbb{R})$ with inner product $\langle f, g\rangle = \int_0^1 f(t)g(t)\,dt$, find an orthogonal basis for $\operatorname{span}(\{1, t, t^2\})$.

**Solution.** We apply the Gram-Schmidt process to the given basis where $\mathbf{v}_1 = 1$, $\mathbf{v}_2 = t$, and $\mathbf{v}_3 = t^2$. First set $\mathbf{w}_1 = 1$. Then

$$\mathbf{w}_2 = \mathbf{v}_2 - \operatorname{proj}_{\mathbf{w}_1}(\mathbf{v}_2) = t - \frac{\int_0^1 1t\,dt}{\int_0^1 1^2\,dt}\,1 = t - \frac{\left[\frac{t^2}{2}\right]_0^1}{[t]_0^1} = t - \tfrac{1}{2},$$

and

$$\begin{aligned}\mathbf{w}_3 = \mathbf{v}_3 - \operatorname{proj}_{\mathbf{w}_1}(\mathbf{v}_3) - \operatorname{proj}_{\mathbf{w}_2}(\mathbf{v}_3) &= t^2 - \frac{\int_0^1 1t^2\,dt}{\int_0^1 1^2\,dt}\,1 - \frac{\int_0^1 (t-\frac{1}{2})t^2\,dt}{\int_0^1 (t-\frac{1}{2})^2\,dt}\,(t - \tfrac{1}{2}) \\ &= t^2 - \frac{\left[\frac{t^3}{3}\right]_0^1}{[t]_0^1} - \frac{\left[\frac{t^4}{4} - \frac{t^3}{6}\right]_0^1}{\left[\frac{t^3}{3} - \frac{t^2}{2} + \frac{t}{4}\right]_0^1}(t - \tfrac{1}{2}) \\ &= t^2 - t + \tfrac{1}{6}.\end{aligned}$$

So an orthogonal basis is $\{1, t - \frac{1}{2}, t^2 - t + \frac{1}{6}\}$. (Via further calculation, we can find an orthonormal basis $\left\{1, 2\sqrt{3}\left(t - \frac{1}{2}\right), 6\sqrt{5}\left(t^2 - t + \frac{1}{6}\right)\right\}$.)

# Orthogonal complement

**Definition.** Given a subspace $X$ of an inner product space $V$, the orthogonal complement of $X$ in $V$, denoted $X^\perp$, is the set $X^\perp = \{\mathbf{y} \in V : \langle \mathbf{y}, \mathbf{x} \rangle = 0 \text{ for all } \mathbf{x} \in X\}$.

**Theorem.** Given a finite-dimensional inner product space $V$, for any $W \leq V$ we have $V = W \oplus W^\perp$.

**Proof.** By Gram-Schmidt, we can construct an orthonormal basis for $W$, say $\{\mathbf{e}_1, \dots, \mathbf{e}_k\}$, and can further build this up into a basis for $V$, say $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ for $n \geq k$. As established in our proof for complementary subspace construction (Lecture 2.3), the space $X = \operatorname{span}(\{\mathbf{e}_{k+1}, \dots, \mathbf{e}_n\})$ is a complementary subspace of $W$. Furthermore, by construction any $\mathbf{v} \in \operatorname{span}(\{\mathbf{e}_{k+1}, \dots, \mathbf{e}_n\})$ is orthogonal with $W$, and any $\mathbf{v} \in W^\perp$ satisfies

$$\mathbf{v} = \sum_{i=1}^{n} \langle \mathbf{e}_i, \mathbf{v} \rangle \mathbf{e}_i = \sum_{i=k+1}^{n} \langle \mathbf{e}_i, \mathbf{v} \rangle \mathbf{e}_i \in X,$$

so in fact $X = W^\perp$. Thus $V = W \oplus W^\perp$.

**Corollary.** Given any subspace $W$ of a finite-dimensional inner product space $V$, we have $(W^\perp)^\perp = W$.

# Properties of the orthogonal complement

**Corollary.** Given any subspace $W$ of a finite-dimensional inner product space $V$, for every $\mathbf{v} \in V$ there is a unique representation $\mathbf{v} = \mathbf{w} + \mathbf{w}'$ where $\mathbf{w} \in W$ and $\mathbf{w}' \in W^\perp$.

**Notation.** With the decomposition above, we write $\mathbf{w} = \mathrm{proj}_W(\mathbf{v})$ and $\mathbf{w}' = \mathrm{proj}_{W^\perp}(\mathbf{v})$. Note that $W = \mathrm{im}(\mathrm{proj}_W)$ and $W^\perp = \ker(\mathrm{proj}_W)$.

**Lemma.** If $\mathbf{v} \in V$, then $\mathbf{v} - \mathrm{proj}_W(\mathbf{v}) \in W^\perp$.

**Proof.** The direct sum decomposition for $\mathbf{v}$ over $W \oplus W^\perp$ is clearly $\mathbf{v} = \mathrm{proj}_W(\mathbf{v}) + (\mathbf{v} - \mathrm{proj}_W(\mathbf{v}))$, where the second summand is $\mathrm{proj}_{W^\perp}(\mathbf{v})$.

**Lemma.** If $\mathbf{w} \in W$, then $\mathrm{proj}_W(\mathbf{w}) = \mathbf{w}$.

**Proof.** The direct sum decomposition for $\mathbf{w}$ over $W \oplus W^\perp$ is clearly $\mathbf{w} = \mathbf{w} + \mathbf{0}$, where the first term is $\mathrm{proj}_W(\mathbf{w})$.

**Lemma.** For all $\mathbf{v} \in V$, we have $||\,\mathrm{proj}_W(\mathbf{v})|| \leq ||\mathbf{v}||$.

**Proof.** Writing the direct sum decomposition for $\mathbf{v}$ over $W \oplus W^\perp$ as $\mathbf{v} = \mathbf{w} + \mathbf{w}'$, we have

$$||\mathbf{v}||^2 = \langle \mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{w}, \mathbf{w} \rangle + \langle \mathbf{w}, \mathbf{w}' \rangle + \langle \mathbf{w}', \mathbf{w} \rangle + \langle \mathbf{w}', \mathbf{w}' \rangle = ||\mathbf{w}||^2 + 0 + 0 + ||\mathbf{w}'||^2.$$

So by positive definiteness, $||\mathbf{v}||^2 \geq ||\mathbf{w}||^2$ and $||\mathbf{v}|| \geq ||\mathbf{w}|| = ||\,\mathrm{proj}_W(\mathbf{v})||$.